Short | Box-Cox gamma sweep | Val ANN10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0689	0.279	0.913	0.853	0.282	1	0.177	0.935	1.08	0	18/18
1 ANN10	0.1	0.0473	0.249	0.958	0.839	0.251	2.81	0.194	0.982	1.19	0	18/18
1 ANN10	0.2	0.0451	0.358	0.945	0.949	0.314	0.973	0.125	0.97	1.04	0	18/18
1 ANN10	0.3	0.0283	0.276	0.667	0.895	0.141	0.898	0.0493	1.42	12.7	0	18/18
1 ANN10	0.5	0.0606	0.584	0.965	0.999	0.175	0.918	0.0938	1	1	0	18/18
1 ANN10	1	0.193	0.903	1.56	1.12	0.801	0.98	0.386	1.17	52.4	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Val ANN10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0629	0.165	0.912	0.879	0.121	0.885	0.0828	0.621	0.844	0	18/18
1 ANN10	0.1	0.0263	0.133	0.725	4.41	0.206	0.908	0.0474	1.03	3.29	0	18/18
1 ANN10	0.2	0.0511	0.215	0.732	0.832	0.267	0.99	0.122	0.712	0.981	0	18/18
1 ANN10	0.3	0.0611	0.339	0.685	0.725	0.213	0.831	0.0716	0.763	1.19	0	18/18
1 ANN10	0.5	0.0301	0.419	0.476	0.596	0.286	0.632	0.0554	0.674	4.08	0	18/18
1 ANN10	1	0.179	0.98	1.53	0.755	1.02	0.929	0.334	5.12	70.7	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Val P-only DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
2 P-only DeepONet10	0.05	0.182	0.779	0.955	0.991	0.267	2.76	0.28	0.998	1	0	18/18
2 P-only DeepONet10	0.1	0.0508	0.0899	0.932	0.966	0.0963	0.953	0.0673	0.972	0.99	0	18/18
2 P-only DeepONet10	0.2	0.0606	0.379	0.884	0.946	0.092	0.615	0.104	1.03	1.38	0	18/18
2 P-only DeepONet10	0.3	0.235	0.514	0.957	10.3	0.236	2.24	0.386	1.14	11.1	0	18/18
2 P-only DeepONet10	0.5	0.324	0.808	2.2	2.4	0.83	1.56	0.443	0.984	3.15	0	18/18
2 P-only DeepONet10	1	0.179	0.637	1.37	2.09	0.698	0.985	0.273	1.07	22.5	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Val P-only DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
2 P-only DeepONet10	0.05	0.0434	0.24	2.38	0.559	0.0875	1.01	0.0504	0.968	1.04	0	18/18
2 P-only DeepONet10	0.1	0.0216	0.0604	0.528	0.522	0.0895	0.253	0.0198	0.438	0.746	0	18/18
2 P-only DeepONet10	0.2	2.11e-03	0.0247	0.168	0.135	0.0351	0.0554	0.0141	0.48	0.621	0	18/18
2 P-only DeepONet10	0.3	0.0127	0.411	0.74	0.454	0.0852	0.212	0.0417	0.975	0.995	0	18/18
2 P-only DeepONet10	0.5	0.353	0.493	1.86	4.16	0.465	2.31	0.239	5.21	109	0	18/18
2 P-only DeepONet10	1	0.767	0.708	16.4	18.4	0.722	4.46	0.494	1	818	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Val ANN10 + P-only DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
3 ANN10 + P-only DeepONet10	0.05	0.1	0.508	0.952	0.938	0.223	1.96	0.231	0.998	1.06	0	18/18
3 ANN10 + P-only DeepONet10	0.1	0.035	0.0815	0.818	0.966	0.0525	0.595	0.0485	0.662	1.01	0	18/18
3 ANN10 + P-only DeepONet10	0.2	0.036	0.145	0.864	0.737	0.106	0.213	0.0667	0.971	1.23	0	18/18
3 ANN10 + P-only DeepONet10	0.3	0.0433	0.445	0.843	0.871	0.145	1.04	0.0843	1.03	1.83	0	18/18
3 ANN10 + P-only DeepONet10	0.5	0.0597	0.525	1.47	1.18	0.437	1.82	0.113	0.958	16.1	0	18/18
3 ANN10 + P-only DeepONet10	1	0.0874	0.593	1.16	1.37	0.6	0.762	0.216	1.32	37.5	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Val ANN10 + P-only DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
3 ANN10 + P-only DeepONet10	0.05	0.0397	0.24	1.41	0.58	0.109	0.663	0.0326	0.701	1.05	0	18/18
3 ANN10 + P-only DeepONet10	0.1	8.90e-03	0.0448	0.282	0.296	0.0716	0.0748	0.0193	0.401	0.496	0	18/18
3 ANN10 + P-only DeepONet10	0.2	1.24e-03	0.015	0.113	0.093	0.0232	0.0461	4.62e-03	0.14	0.332	0	18/18
3 ANN10 + P-only DeepONet10	0.3	0.012	0.159	0.414	0.478	0.0579	0.0728	0.0278	0.748	0.877	0	18/18
3 ANN10 + P-only DeepONet10	0.5	0.0577	0.458	1.33	0.928	0.414	1.07	0.114	1.62	28.8	0	18/18
3 ANN10 + P-only DeepONet10	1	0.371	0.68	2.64	2.95	1.02	1.15	0.66	1.8	241	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Val T/P DeepONet10: common ICs [0, 10, 14, 17, 19, 24, 33, 46, 63]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
4 T/P DeepONet10	0.05	0.222	1.13	1	1	0.67	1	0.399	0.999	1	0	18/18
4 T/P DeepONet10	0.1	0.0909	0.534	1	0.813	0.243	0.957	0.108	0.999	1	0	18/18
4 T/P DeepONet10	0.2	0.106	1	0.709	1	0.319	0.302	0.164	0.998	1	0	18/18
4 T/P DeepONet10	0.3	0.0336	0.58	0.42	0.237	0.0864	0.149	0.0673	0.723	0.776	0	18/18
4 T/P DeepONet10	0.5	0.464	0.723	6	2.79	0.687	1.23	0.276	26.7	28.9	0	13/18
4 T/P DeepONet10	1	0.261	3.13	1	1.57	0.698	0.549	0.379	0.966	30.8	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Val T/P DeepONet10: common ICs [0, 10, 14, 17, 19, 24, 33, 46, 63]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
4 T/P DeepONet10	0.05	0.364	0.921	9.99	2.52	0.825	3.58	0.335	1.16	1.41	0	18/18
4 T/P DeepONet10	0.1	0.0367	0.478	0.617	0.368	0.0907	0.672	0.052	0.837	1	0	18/18
4 T/P DeepONet10	0.2	9.64e-03	0.128	0.15	0.16	0.0224	0.0542	0.0114	0.242	0.264	0	18/18
4 T/P DeepONet10	0.3	0.0171	0.374	0.166	0.187	0.0333	0.0256	0.0223	0.218	0.258	0	18/18
4 T/P DeepONet10	0.5	0.11	1.52	0.999	1	0.366	0.319	0.163	0.675	0.991	0	18/18
4 T/P DeepONet10	1	0.25	3.41	1.23	1.67	0.995	0.627	0.367	0.871	39.3	0	14/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Val ANN10 + T/P DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
5 ANN10 + T/P DeepONet10	0.05	0.195	0.332	1.68	0.981	0.591	2.49	0.222	1.59	1.59	0	18/18
5 ANN10 + T/P DeepONet10	0.1	0.013	0.0698	0.739	0.598	0.0845	0.899	0.0764	0.842	0.882	0	18/18
5 ANN10 + T/P DeepONet10	0.2	0.0241	0.255	0.444	0.673	0.169	0.18	0.0741	0.986	1	0	18/18
5 ANN10 + T/P DeepONet10	0.3	9.41e-03	0.115	0.312	0.326	0.0766	0.158	0.0215	0.812	0.895	0	18/18
5 ANN10 + T/P DeepONet10	0.5	0.0918	0.804	1.04	2.57	0.734	1.12	0.185	1.18	1.76	0	18/18
5 ANN10 + T/P DeepONet10	1	0.157	0.738	2.26	1.77	0.822	0.578	0.282	0.983	27.1	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Val ANN10 + T/P DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
5 ANN10 + T/P DeepONet10	0.05	0.129	0.265	0.986	0.938	0.123	0.997	0.236	1.01	1.24	0	18/18
5 ANN10 + T/P DeepONet10	0.1	0.0228	0.119	0.551	0.456	0.0646	0.607	0.0322	0.588	0.696	0	18/18
5 ANN10 + T/P DeepONet10	0.2	5.63e-03	0.0718	0.196	0.436	0.0642	0.141	0.0332	0.839	0.958	0	18/18
5 ANN10 + T/P DeepONet10	0.3	3.23e-03	0.0335	0.144	0.2	0.0292	0.0647	0.0105	0.358	0.303	0	18/18
5 ANN10 + T/P DeepONet10	0.5	0.0422	0.6	0.893	0.725	0.212	0.267	0.0846	0.704	0.835	0	18/18
5 ANN10 + T/P DeepONet10	1	0.313	0.991	2.96	2.73	0.955	1.04	0.427	9.22	59	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Val T/P/ϕ DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 33, 41, 46, 48, 63, 71]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
6 T/P/ϕ DeepONet10	0.05	0.384	0.561	5.08	1.03	0.375	4.23	0.389	4.14	2.27	0	18/18
6 T/P/ϕ DeepONet10	0.1	0.136	0.465	1	0.802	0.245	1	0.122	0.999	1	0	18/18
6 T/P/ϕ DeepONet10	0.2	0.0229	0.222	0.5	0.452	0.169	0.308	0.04	0.653	0.68	0	18/18
6 T/P/ϕ DeepONet10	0.3	0.362	0.878	1	1.45	0.278	1.3	0.3	1	2.17	0	18/18
6 T/P/ϕ DeepONet10	0.5	0.168	0.669	1.95	0.99	0.671	1	0.294	1	1	0	17/18
6 T/P/ϕ DeepONet10	1	0.133	0.975	3.38	1.85	0.703	0.543	0.186	1.08	10.3	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Val T/P/ϕ DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 33, 41, 46, 48, 63, 71]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
6 T/P/ϕ DeepONet10	0.05	0.268	0.414	2.31	0.968	0.409	3.36	0.207	1.23	1.33	0	17/18
6 T/P/ϕ DeepONet10	0.1	0.331	0.349	4.44	0.858	0.602	6.75	0.359	1.01	1.01	0	18/18
6 T/P/ϕ DeepONet10	0.2	0.035	0.39	0.96	0.743	0.123	0.243	0.0573	0.996	1.06	0	18/18
6 T/P/ϕ DeepONet10	0.3	0.0945	0.663	1.02	0.544	0.294	0.181	0.0755	0.794	0.972	0	18/18
6 T/P/ϕ DeepONet10	0.5	0.13	0.668	0.918	0.913	0.232	0.783	0.0902	0.952	1.06	0	18/18
6 T/P/ϕ DeepONet10	1	0.531	1.03	2.83	2.22	0.932	1.44	0.342	4.08	69.6	0	16/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Short | Box-Cox gamma sweep | Val ANN10 + T/P/ϕ DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.142	0.385	0.988	0.977	0.663	1	0.378	1.29	2.38	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0442	0.103	0.724	0.345	0.105	0.942	0.0906	0.842	0.865	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.013	0.138	0.369	0.285	0.0847	0.268	0.0388	0.808	0.929	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0751	0.361	0.988	0.993	0.153	0.926	0.141	1	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0582	0.678	0.854	0.866	0.631	0.757	0.173	0.981	0.974	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	1	0.186	0.989	2.76	1.24	0.848	0.468	0.211	1.11	15.3	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Long | Box-Cox gamma sweep | Val ANN10 + T/P/ϕ DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.359	0.323	4.7	0.959	0.145	5.27	0.327	1.18	1.65	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.191	0.223	2.81	0.506	0.154	4.98	0.169	0.92	0.991	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0208	0.168	0.814	0.4	0.0553	0.149	0.0289	0.35	0.559	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0839	0.308	0.999	0.706	0.172	0.499	0.0829	0.779	0.843	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0714	0.267	0.957	0.706	0.204	0.697	0.0769	0.945	0.995	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	1	0.383	0.839	5.01	2.83	0.82	1.15	0.329	3.93	56.6	0	18/18

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.